

Kyu Hyun Lee

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Academic Appointments

Postdoctoral Scholar, University of California, San Francisco

2020–present

Advisor: Loren M. Frank

Education

California Institute of Technology

2013–2020

Ph.D., Biochemistry and Molecular Biophysics option

Advisor: Markus Meister

Pomona College

2007–2011

B.A., Chemistry (minor: Mathematics), *magna cum laude*

Research Funding and Fellowships

NIH/NEI Pathway to Independence Award (K99/R00)

2025–present

Principal Investigator • K99EY036953

Helen Hay Whitney Foundation Postdoctoral Fellowship

2022–2025

Tianqiao and Chrissy Chen Institute Graduate Innovator Grant

2018

Caltech-Amgen Graduate Fellowship

2014–2015

Selected Research Outputs

Manuscript in preparation

Lee, K., Adenekan, P., Gao, F., Zhou, J., Hernandez, J., Yorita, A., Haque, R., Kay, K., Scanziani, M., and Frank, L.M. Vision gain-modulates a nonvisual abstract representation in primary visual cortex.

Book

Meister, M., **Lee, K.**, and Portuguese, R. (2025). *Mathematics in Biology*. MIT Press.

Peer-reviewed publications

- Guidera, J.A., Gramling, D.P., Comrie, A.E., Joshi, A., Tseng, S.Y., Denovellis, E.L., Smyth, C.N., **Lee, K.**, Zhou, J., Thompson, P., Hernandez, J., Yorita, A., Haque, R., Kirst, C., and Frank, L.M. (2026). Regional specialization in prefrontal cortex manifests in the reliability of task progression codes. *Neuron* 114(6), 1152–1161.e8.
- Lee, K.**^{*}, Denovellis, E.^{*}, Ly, R., Magland, J., Soules, J., Comrie, A.E., Gramling, D.P., Guidera, J.A., Nevers, R., Adenekan, P., Brozdowski, C., Bray, S., Monroe, E., Bak, J., Coulter, M., Sun, X., Broyles, E., Shin, D., Chiang, S., Holobetz, C., Tritt, A., Rübel, O., Nguyen, T., Yatsenko, D., Chu, J., Kemere, C., Garcia, S., Buccino, A., and Frank, L.M. (2025). Spyglass: a framework for reproducible and shareable neuroscience research. *eLife* 14, RP108089. Reviewed Preprint.
- Lee, K.**^{*}, Ni, Y.^{*}, Colonell, J., Karsh, B., Putzeys, J., Pachitariu, M., Harris, T.D., and Meister, M. (2021). Electrode pooling can boost the yield of extracellular recordings with switchable silicon probes. *Nature Communications* 12, 5245.
- Lee, K.**, Tran, A., Turan, Z., and Meister, M. (2020). The sifting of visual information in the superior colliculus. *eLife* 9, e50678.

^{*}Equal contribution.

Additional Research Outputs

Additional peer-reviewed publications

5. **Lee, K.**, Humbarger, S., Bahnvadia, R., Sazinsky, M.H., and Crane, E.J., III (2014). Characterization of the mechanism of the NADH-dependent polysulfide reductase (Npsr) from *Shewanella loihica* PV-4: Formation of a productive NADH-enzyme complex and its role in the general mechanism of NADH and FAD-dependent enzymes. *Biochimica et Biophysica Acta - Proteins and Proteomics* 1844(9), 1708–1717.
6. Warner, M.D., Lukose, V., **Lee, K.**, Lopez, K., Sazinsky, M.H., and Crane, E.J., III (2011). Characterization of an NADH-dependent persulfide reductase from *Shewanella loihica* PV-4: Implications for the mechanism of sulfur respiration via FAD-dependent enzymes. *Biochemistry* 50(2), 194–206.

Peer-reviewed conference proceeding

Hermiz, J., Joseph, E., **Lee, K.**, Baldacci, I.A., Chung, J.E., Frank, L.M., Bouchard, K.E., and Denes, P. (2021). The impact of reducing signal acquisition specifications on neuronal spike sorting. *2021 43rd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)*, 5914–5918.

Patent

Meister, M., **Lee, K.**, and Ni, Y. (2024). Methods and systems for electrode pooling. U.S. Patent No. 12,097,033.

Invited Talks and Presentations

Navigating Educational Trajectories in Neuroscience (NIH R25 Program) University of Oregon	2024, 2025
Dynamic Brain Circuits Cluster Databinge University of British Columbia, Djavad Mowafaghian Centre for Brain Health	2024
“Reproducible Workflows for Collaborative Neuroscience” symposium BRAIN Initiative Meeting, NIH	2023
Spike Sorting Workshop Janelia Research Campus	2018
Junior Scientist Workshop on Neural Circuits and Behavior Janelia Research Campus	2017

Honors and Awards

Phi Beta Kappa	2011
Robert Rowan III Memorial Award	2011

Research Mentoring

Denisse Morales-Rodriguez , Graduate Student, UCSF Optogenetic manipulation of mPFC during sleep.	2024–present
Scott Gao , Research Associate, UCSF Hippocampus–visual cortex interactions.	2024–present
Philip Adenekan , Research Associate, UCSF Hippocampus–visual cortex interactions.	2022–2024
Yangsun Hwang , Rotation Student, UCSF Chronic tracking of neurons recorded with polymer probes.	2023
Roshanak Gonzalez , Rotation Student, UCSF Testing an optoelectronic neural probe.	2022
Daryl Lawrence , Rotation Student, UCSF Denoising signals from a neural-recording amplifier chip.	2021

Teaching Experience

Division of Biology and Biological Engineering, California Institute of Technology	
Principles of Neuroscience (Fall terms)	2018
Tools of Neurobiology (Fall terms)	2016
Mathematics in Biology (Fall terms)	2015
Brain Circuits (Winter terms)	2014